

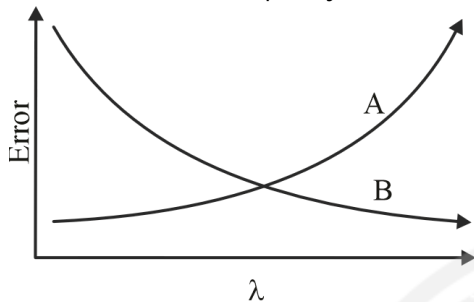
DS&AI

Machine Learning

DPP: 1

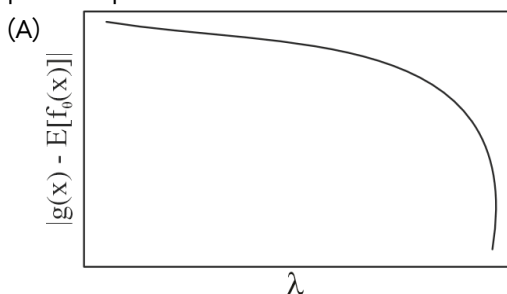
Multi Layer Perceptron

Q1 Assuming a regularization penalty of the form $\lambda R(q)$. Complete the following illustration. Note that the x-axis is the regularization parameter λ and not the model complexity.

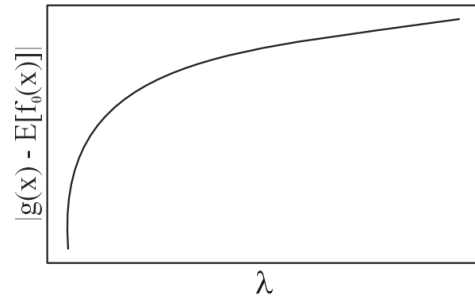


- (A) Is the test Error and (B) is the error due to $(\text{Bias})^2$.
 (B) Is the error due to Model variance and (B) is the Training Error.
 (C) Is the error due to Model variance and (B) is the error due to $(\text{Bias})^2$.
 (D) Is the error due to $(\text{Bias})^2$ and (B) is the error to model Variance.

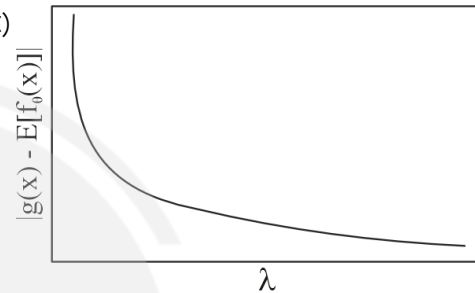
Q2 Which of the following could plausibly be a plot of the magnitude (i.e., absolute value) of model bias as a function of λ if we use multiple linear regression with exponential regularization? Assume that values on the x-axis are linear from 0 asymptotically large λ . Circle the letter corresponding to the most plausible plot.



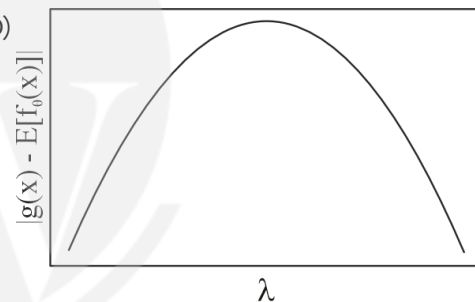
(B)



(C)



(D)



Q3 Ishani trains two model on the same data with the same features: Model A and Model B. Information on both models is given below:

Model	Bias Squared	Model Variance
A	2	12
B	4	1

Which of the following statements are true?

Select at that apply.

- (A) Model A will overfit more than Model B for this dataset
 (B) If both models used LASSO Regression, Model B had the higher λ .
 (C) Model A could have used Ridge Regression with $\lambda > 0$ while Model B used OLS.
 (D) Model B must be larger weights than Model A.



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Q4 Matthew decides to try a new method. He first comes up with a model $h(x)$, which has a bias of B and a variance of V on the original dataset. Next, he builds a new model $f(x)$ using this process:

1. Obtain n independent, random samples from the original dataset.
2. For each dataset, train a new version of $h(x)$ for each i from 1 to n , where $h_1(x)$ was the $h(x)$ model trained on the first dataset and $h_n(x)$ was the $h(x)$ model trained on the n th dataset.
3. Finally, define $f(x)$ as $f(x) = \sum_{i=1}^n h_i(x)$

What is the model's bias squared of the $f(x)$ model in terms of B and n ? Grading will be done based on the work you shown in the box below.

- (A) B^2 (B) $\frac{1}{n}B^2$
(C) $\left(\frac{B}{n}\right)^2$ (D) nB^2

Q5 If we assume that the data is being generated from a source with no noise $y = w^T x$, choosing a classifier $f^*(x) = w^T x$ using OLS will make which of the following quantities zero?

- (A) Only Bias
(B) Only Variance
(C) Both Bias and Variance
(D) None of the above

Q6 Consider a Multi-Layer Perceptron (MLP) model with one hidden layer and one output layer. The hidden layer has 10 neurons, and the output layer has 3 neurons. The input to the MLP is a 5-dimensional vector. Each neuron is connected to every neuron in the previous layer, and a bias term is included for each neuron. The activation function used is the sigmoid function. Calculate the total number of trainable parameters in this MLP model.

Q7 Consider the following binary classification dataset:

- (x_1, x_2, x_3, y)
(0, 0, 1, +1)
(0, 1, 1, -1)
(1, 0, 1, -1)

x_3 is used for the bias term.

Suppose we initialize the weight vector for a linear classifier to $w = (0, 0, 0)$.

What are the weights after one pass through the data in the order listed above?

Q8 Given the following training points for classification, where each point is represented as (x_1, x_2, y)

- (1, 0, +)
(1, 1, +)
(0, 1, -)

Perform one pass of the perceptron algorithm on this data, starting from a weight vector initialized at $w = [0, 0]$ and using the decision rule $w^T x > 0$, (i.e., classifying as positive if the source is greater than 0.)

Assume on bias term is used.

What is the final weight vector w after this pass?

- (A) [1, 1] (B) [2, 1]
(C) [1, 0] (D) [0, 1]

Q9 Assume that you are given observations (x, y) $\hat{y} \in \{\pm 1\}$ in the following order:

Instance	1	2	3	4	5	6	7	8
Label y	+1	-1	+1	-1	+1	-1	+1	+1
Data (x_1, x_2)	(10, 10)	(0, 0)	(8, 4)	(3, 3)	(4, 8)	(0.5, 0.5)	(4, 3)	(2, 5)

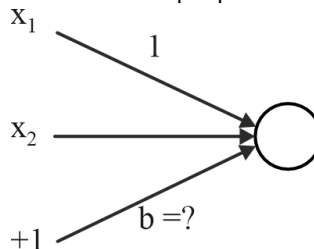
Show the action of the perceptron algorithm for the above sequence of observations. We start with an initial set of weights $w = (1, 1)$ and bias $b = 0$.

- (A) $w_{\text{new}} = [-2 \ 1.5 \ 5.5]$
(B) $w_{\text{new}} = [-2 \ 1.8 \ 5.5]$
(C) $w_{\text{new}} = [-2 \ 1.5 \ 8.5]$
(D) $w_{\text{new}} = [-2 \ 1.5 \ 5.9]$

Q10 Suppose we want to implement logical OR using perceptron.

For the given decision rule, what will be the bias term if weights are given as $w_1 = w_2 = 1$.

There are multiple possibilities, gives anyone.



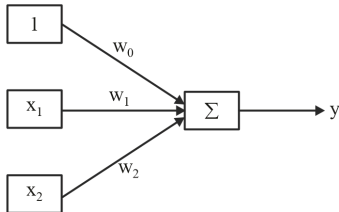
OR		
X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	1



$$y = \begin{cases} 0, & \text{if } w \cdot x + b \leq 0 \\ 1, & \text{if } w \cdot x + b > 0 \end{cases}$$

- (A) $-1 < b < 3$ (B) $-1 < b < 0$
 (C) $-1 < b < 1$ (D) $-1 < b < 2$

Q11 Consider the following perceptron, for which the inputs are the always 1 features and two binary features $x_1 \in \{0,1\}$ and $x_2 \in \{0,1\}$. The output $y \in \{0,1\}$.



$$y = \begin{cases} 1, & \text{if } (w_1 \cdot x_1 + w_2 \cdot x_2) > 0 \\ 0, & \text{otherwise} \end{cases}$$

Which one(s) of the following choices for the weight vector $[w_0 \ w_1 \ w_2]$ can classify y as $y = (x_1 \text{ XOR } x_2)$? XOR is the logical exclusive or operation, which equals to zero when x_1 equals to x_2 and to one when x_1 is different from x_2 .

- (A) $[1 \ 1 \ 0]$
 (B) $[1 \ 1 \ 0]$
 (C) $[-2 \ 1 \ 1.5]$
 (D) Any weight that satisfy $(-w_1 - w_2) < w_0 < \min(0, -w_1, -w_2)$
 (E) No weight can compute the XOR logical relation.

Q12 Consider a Perceptron that needs to classify binary input data correctly. The dataset includes the following input-output pairs, where each input has three binary values, and the output is a binary classification label:

(111, 1) (011, 1)
 (110, 1) (010, 0)
 (000, 0)

The rule for classification is: the output should be 1 if at least two of the three input are 1; otherwise, the output should be 0.

Can you determine a suitable set of weights and a threshold value for the perceptron so that it correctly classifies this dataset?

- (A) $x_1 + x_2 + x_3 > 2$ (B) $x_1 + x_2 + x_3 > 7$
 (C) $x_1 + x_2 + x_3 > 6$ (D) $x_1 + x_2 + x_3 > 4$

Q13

Which of the following is/are true about the perceptron classification?

- (A) It can learn a OR function
 (B) It can learn a XOR function
 (C) The obtained separating hyperplane depends on the order in which the points are presented in the training process.
 (D) For a linearly separable problem, there exists some initialization of the weights which might lead to non-convergent cases.

Q14 Consider a binary classification problem.

Suppose I have trained a model on a linearly separable training set, and now I get a new labeled data point which is correctly classified by the model, and far away from the decision boundary. If I now add this new point to my earlier training set and re-train, in which cases is the learnt decision boundary likely to change?

- (A) When my model is a perceptron.
 (B) When my model is logistic regression
 (C) When my model is an SVM
 (D) None of these

Q15 The perceptron algorithm will converge:

- (A) If the data is linearly separable
 (B) As long as you initialize θ to all 0's
 (C) Even if the data is linearly inseparable
 (D) Always

Q16 Leave-one-out cross-validation (LOOCV) is a special case of k-fold cross validation where:

- (A) The training set contains all but one sample, and the remaining samples are used for testing.
 (B) The training set contains only one sample, and the remaining sample is used for testing.
 (C) The training set contains exactly one sample from each class, and the remaining samples are used for testing.
 (D) The training set contains one sample from each fold, and the remaining folds are used for testing.

Q17 Why is it important to use a different test set to evaluate the final performance of the model, rather than the validation set used during model selection?

- (A) The model may have overfit to fit the validation set
 (B)



The test set is a better representation of new, unseen data

- (C) Both a and b
(D) None of the above

Q18 Under which of the following conditions is k-fold cross-validation the same as leave-one-out cross-validation?

- (A) The training set and test set have the same number of examples
(B) The training set and tuning set have the same number of examples
(C) $k = 1$
(D) $k = n$, where n is the total number of examples
(E) None of the above

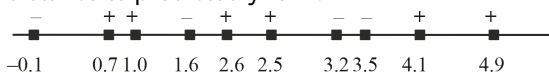
Q19 Suppose you are using a Majority Classifier on the following training set containing 10 examples where each example has one real-valued feature, x , and a binary class label, y , with value 0 or 1. Define this Majority Classifier to predict the class label that is in the majority in the training set, regardless of the input value. In case of ties, predict class 1.



What is the training set accuracy?

- (A) 0%
(B) 10%
(C) 50%
(D) 90%
(E) 100%

Q20 Consider the following dataset with one real-value input x and one binary output y . We are going to use k-NN with unweighted Euclidean distance to predicted y for x .



X	Y
-0.1	-
0.7	+
1.0	+
1.6	-
2.0	+
2.5	+
3.2	-

3.5	-
4.1	+
4.9	+

What is the leave-one-out cross validation error of 1-NN on this dataset? Give your answer as the number of misclassifications.

- (A) 3/10
(B) 2/10
(C) 5/10
(D) 6/10

Q21 To prevent overfitting of our linear model, we would apply regularization on the model. Which of the following dataset should we evaluate using our model in order to decide the right amount of regularization?

- (A) Train
(B) Validation
(C) Test
(D) All of the above

Q22 What is cross-validation not used for?

- (A) To evaluate the performance of a machine learning model on unseen data.
(B) To select a model's hyperparameters
(C) To determine the generalization of a machine learning model.
(D) To train multiple machine models on different datasets.

Q23 Consider a single perceptron with sign activation function. The perceptron is represented by weight vector $[0.4 \ -0.3 \ 0.1]^T$ and a bias $\theta = 0$. If the input vector to the perceptron is $X = [0.2 \ 0.6 \ 0.5]$ then the output of the perceptron is :

Q24 Consider a Multi-Layer Perceptron (MLP) model with one hidden layer and one output layer. The hidden layer has 12 neurons, and the output layer has 5 neurons. The input to the MLP is a 6-dimensional vector. Each neuron is fully connected to the previous layer, and a bias term is included for each neuron. The activation function used is ReLU for the hidden layer and Softmax for the output layer. Calculate the total number of trainable parameters in this MLP model.

Q25 Consider an MLP model with three hidden layers and one output layer. The first hidden layer has 20 neurons, the second has 12 neurons, the third has 8 neurons, and the output layer has 3



neurons. The input to the MLP is a 10-dimensional vector. Each neuron is connected to all neurons in the previous layer, and each neuron includes a bias term.

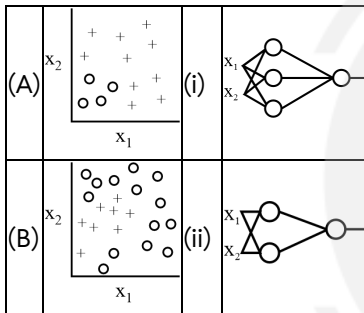
Determine the total number of trainable parameters in this MLP model.

- Q26** Consider a single-layer perceptron with a sign activation function. The perceptron is represented by the weight vector $[0.5, -0.2, 0.3]^T$ and bias $q = 5$. If the input vector to the perceptron is:

$$X = [0.4, 0.7, 0.2]$$

Then, determine the output of the perceptron.

- Q27** Recall that the output of a perceptron is 0 or 1. For each of the two following data sets, select the perceptron network with the fewest nodes that will separate the classes. You can use the same network more than once



- (A) (i)-A; (ii)-B (B) (i)-B; (ii)-A
(C) (i)-A; (ii)-A (D) (i)-B; (ii)-B

- Q28** Let's say you have a MLP network which is trained on 5-dimensional input data. The network has 3 hidden layers, with 50, 30, and 20 nodes respectively. The network is trained to predict a 10-way classification task using softmax activation. If we ignore bias terms, how many weights does this neural network have?

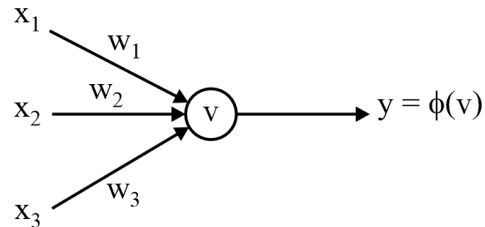
- (A) 2,550 (B) 1,500,000
(C) 115 (D) 30,000

- Q29** Consider the unit shown on Figure. Suppose that the weights corresponding to the three inputs have the following values

$$w_1 = 2$$

$$w_2 = -4$$

$$w_3 = 1$$



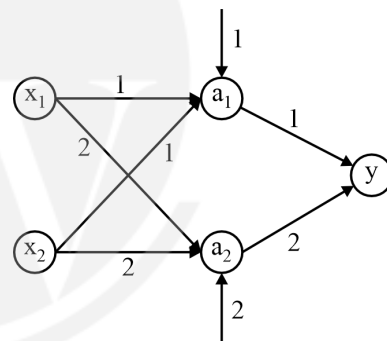
and the activation of the unit is given by the step-function:

$$\phi = \begin{cases} 1 & v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

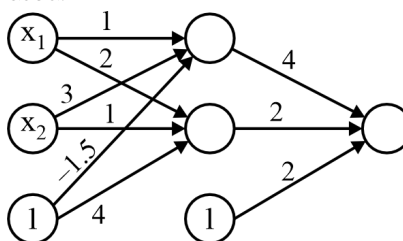
Calculate what will be the sum of output value y of the unit for each of the following input patterns:

Pattern	P ₁	P ₂	P ₃	P ₄
x_1	1	0	1	1
x_2	0	1	0	1
x_3	0	1	1	1

- Q30** What is the output(round to 2 decimal points) of the network for $x_1 = 1, x_2 = 1$? Assume sigmoid is being at a_1, a_2 and at y no activation function is used.



- Q31** What is the output of the network for $x_1 = 1, x_2 = 2$. Assume at every level identity function is used.



- Q32** Consider an MLP model with two hidden layers and one output layer. The first hidden layer has 8 neurons, the second hidden layer has 5 neurons, and the output layer has 3 neurons. The input to the MLP is a 4-dimensional vector. Each neuron is fully connected to every neuron in the previous layer, and a bias term is included for each neuron.



Calculate the total number of trainable parameters in this MLP model.

- Q33** What is the function of the hidden layers in a multilayer perceptron (MLP)?
- (A) Hidden layers perform feature extraction and transformation
 - (B) Hidden layers provide direct access to the input data
 - (C) Hidden layers perform the final classification or regression task

(D) Hidden layers are not necessary in an MLP

- Q34** Consider the following statements:
 S_1 : The Bias-Variance Trade-Off helps in balancing the complexity of the model and its performance on unseen data.
 S_2 : A Multi-Layer Perceptron (MLP) can only approximate linear decision boundaries.
Which of the above is true?
- (A) Only S_1
 - (B) Only S_2
 - (C) Both S_1 and S_2
 - (D) None of these

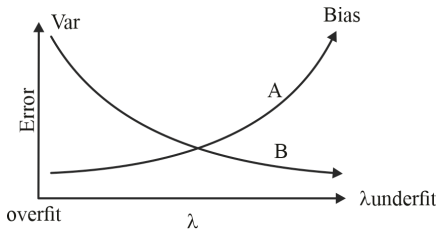


Answer Key

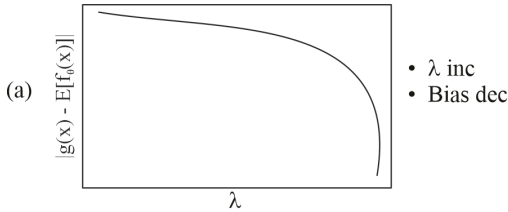
Q1 (D)
Q2 (C)
Q3 (A)
Q4 (B)
Q5 (A)
Q6 93~93
Q7 1~1
Q8 (A)
Q9 (A)
Q10 (B)
Q11 (E)
Q12 (A)
Q13 (A, C)
Q14 (D)
Q15 (A)
Q16 (A)
Q17 (C)

Q18 (D)
Q19 (C)
Q20 (A)
Q21 (B)
Q22 (A)
Q23 -1
Q24 149~149
Q25 603
Q26 +1
Q27 (B)
Q28 (A)
Q29 0
Q30 0
Q31 40~40
Q32 103~103
Q33 (A)
Q34 (A)

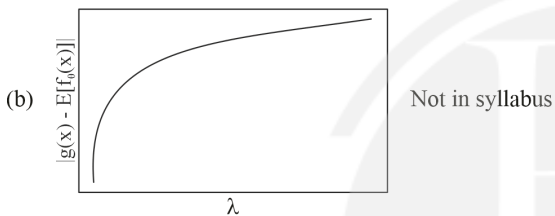
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Q1 Text Solution:

- λ inc: regularization inc
- underfit

Q2 Text Solution:

- λ inc
- Bias dec

**Q3 Text Solution:**

Model	Bias Squared	Model Variance
A	2	12
B	4	1

over fit
under fit

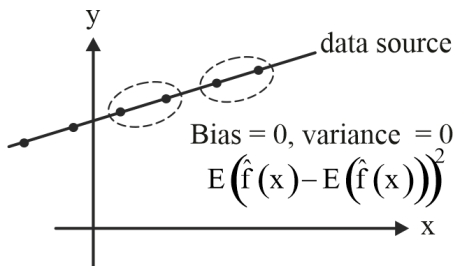
- Model A will overfit more than Model B for this dataset

Q4 Text Solution:

$$\frac{1}{n} B^2$$

Q5 Text Solution:

- Data $\rightarrow$ no noise

**Q6 Text Solution:**

$$\begin{array}{c} 10 \\ \downarrow \\ 50 \\ \downarrow \\ 5 \quad \frac{50}{10} \quad 30 \quad \frac{3}{3} \\ \text{I/P} \quad \text{HD} \quad \text{O/P} \end{array} \Rightarrow 93$$

Q7 Text Solution:

	x_1	x_2	Padding		
learning Rate=1	1	0	0	1	+1
	2	0	1	1	-1
	3	1	1	1	-1

- we start with $w = 000$

$$w = \begin{bmatrix} n \\ w_1 \\ w_2 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

(1)

$$y_i (w^T x_i + b) = 1 (0 + 0 + 0) = 0 \Rightarrow \text{error}$$

$$w_{\text{new}} = w_{\text{old}} + \eta y_i x_i \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$(2) y_i (w_i x_i + b) = 1 (0 + 1) = 1 \Rightarrow \text{error}$$

$$w_{\text{new}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 1 \cdot (-1) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$w_{\text{new}} \Rightarrow \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

(3)

$$y_i (w x_i + b) = -1 (0 + 0 + 0) = 0 \Rightarrow \text{error}$$

$$w_{\text{new}} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

$$w_{\text{new}} \Rightarrow \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$$

Q8 Text Solution:

no bias

$$(1) y_i (w x_i) = 1 \times 0 = 0 \Rightarrow \text{error}$$

$$w_{\text{new}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \cdot (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow w_{\text{new}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(2) y_i (w x_i) = +1 (1) = +1 \Rightarrow \text{no error}$$

$$(3) y_i (w x_i) = -1 (0) = 0 \Rightarrow \text{error}$$

$$w_{\text{new}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$w_{\text{new}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Q9 Text Solution:

$$1. y_i (wx_i + b) = 1 (20 + 10) = +20$$

→ no error

2.

$$y_i (wx_i + b) = (-1) (0 + 0) = 0$$

$$= \text{error} \quad w_{\text{new}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$3. y_i (wx_i + b) = 1 (12 - 1) = +11$$

⇒ no error

$$4. y_i (wx_i + b) = (-1) (5) = -5 \Rightarrow \text{error}$$

$$\Rightarrow w_{\text{new}} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + (-1) \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$$

$$5. y_i (wx_i + b) < 0 \text{ error}$$

$$w_{\text{new}} = \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix} + (1) \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}$$

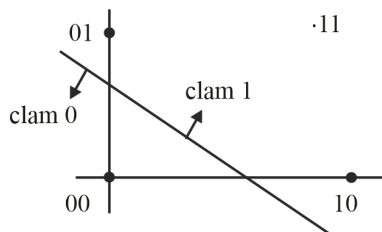
$$6. y_i (wx_i + b) = (-1) (1 + 3 - 1) < 0$$

⇒ error

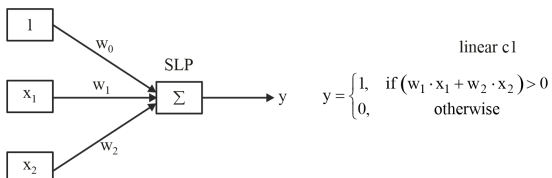
$$w_{\text{new}} = \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} + (-1) \begin{pmatrix} 1 \\ .5 \\ .5 \end{pmatrix} = \begin{pmatrix} -2 \\ 1.5 \\ 5.5 \end{pmatrix}$$

$$w_{\text{new}} \Rightarrow \begin{bmatrix} -2 \\ 1.5 \\ 5.5 \end{bmatrix}$$

Q10 Text Solution:



Q11 Text Solution:



Q12 Text Solution:

- $x_1 + x_2 + x_3 \geq 2 \Rightarrow \text{clam 1}$
- $x_1 + x_2 + x_3 < 2 \Rightarrow \text{clam 0}$

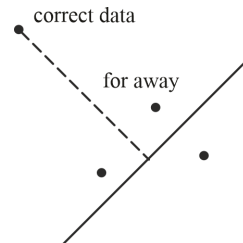
Three input x_1, x_2, x_3

$$\bullet x_1 + x_2 + x_3 \geq 2.$$

Q13 Text Solution:

- It can learn a OR function
- The obtained separating hyperplane depends on the order in which the points are presented in the training process.

Q14 Text Solution:



$$\text{Max } \pi P(x)^{y_i} (1 - P(x))^{1-y_i}$$

Cam 1 point for aways $P(x)=1$

$$P(x) = \frac{1}{1 + e^{-(wx_i)}}$$

$$wx_i \gg 0 \quad P(x) = 1$$

Q15 Text Solution:

- The perceptron algorithm will converge:

Q16 Text Solution:

- The training set contains all but one sample, and the remaining samples is used for testing-
- It's a special case of **k-fold cross-validation** where **k = number of data points (n)**.

Q17 Text Solution:

. A. The model may have overfit to the validation set

- During training, the **validation set** is used to **tune hyperparameters** and **select the best model**.
- If you keep evaluating on the same validation set, you risk **indirectly overfitting** to it.
- That means your model might perform well on the validation set, but not generalize to truly unseen data.

B. The test set is a better representation of new, unseen data

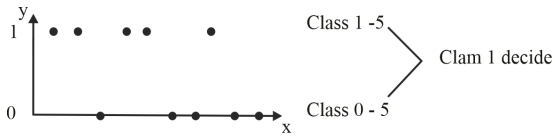
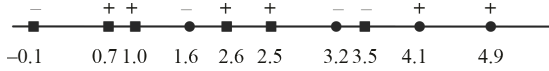
- The **test set** is supposed to simulate **real-world data** — data the model has never seen in any way.
- It gives an **unbiased estimate** of the model's performance after all tuning is done.



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Q18 Text Solution:

- $k = n$, where n is the total number of examples

Q19 Text Solution:**Q20 Text Solution:**

3/10 doc

Q21 Text Solution:

Regularization is used to prevent overfitting by adding a penalty to large weights in a linear model (e.g., L1 or L2 regularization).

Q22 Text Solution:

- To evaluate the performance of a machine learning model on unseen data.
- The select a model's hyperparameters
- The determine the generalization of a machine learning model.

Q23 Text Solution:

Input: $(.4x \cdot 2 + .6x - .3 + .5x \cdot 1)$

$.08 - .18 + .05 \Rightarrow -.05$

Output = -1

Q24 Text Solution:

6×12	12×5	$+12 + 5$
6	12	5
I/P	HD	O/P

= 149

Q25 Text Solution:

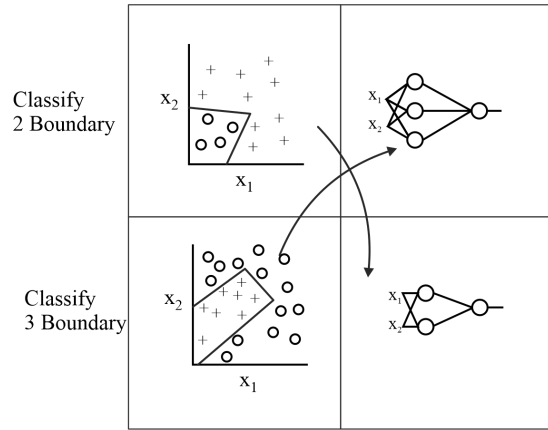
200	240	96	24
10	20	12	8
	20	12	8

= 603

Q26 Text Solution:

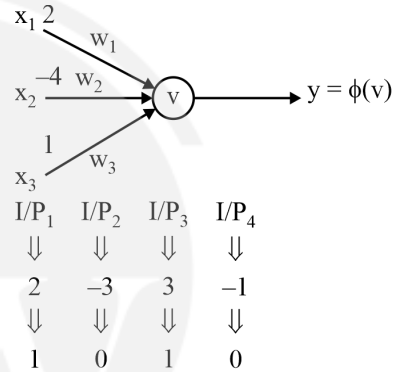
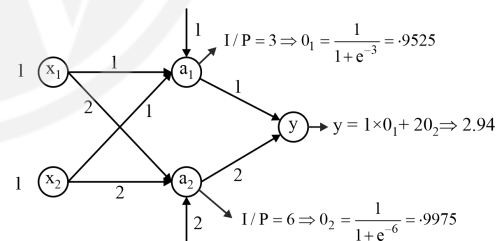
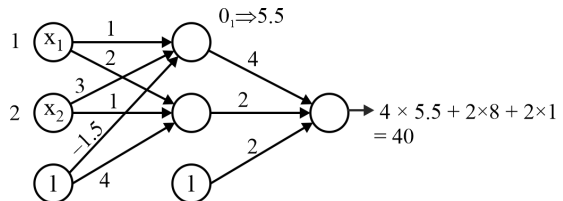
Input = $(.2 - .14 + .06 \cdot + .5) = .5$

Input > 0, Output $\rightarrow +1$

Q27 Text Solution:**Q28 Text Solution:**

250	1500	600	200
5	50	30	20

10 output
 $250 + 1500 + 600 + 200 = 2550$

Q29 Text Solution:**Q30 Text Solution:****Q31 Text Solution:****Q32 Text Solution:**

32	40	15
4	8	5
	8	5

= 103

Q33 Text Solution:

- Hidden layers perform feature extraction and transformation =
- Their main function is to **transform the input data** into more **useful internal representations**, making it easier for the output layer to perform classification or regression.
- This transformation allows the model to **learn complex patterns** and **nonlinear relationships**.

Q34 Text Solution:

- The **Bias-Variance Trade-Off** is a fundamental concept in machine learning.
- **High bias** = model too simple → underfitting.
- **High variance** = model too complex → overfitting.
- A good model finds the **right balance** to perform well on unseen (test) data

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